

3	(a)	State the <i>principle of superposition</i> .	
			
			
			
			
			
			[2]
		Solution:	
		The principle of superposition states that when two or more waves of the same nature meet at a point,	1
		the resultant displacement is the vector sum of the individual displacements due to each wave at that point.	1

- (b) Interference effects are produced on a viewing screen as a result of interference of the direct waves from a 500 nm electromagnetic wave source S and reflected waves from the mirror as shown in Fig. 3.1.

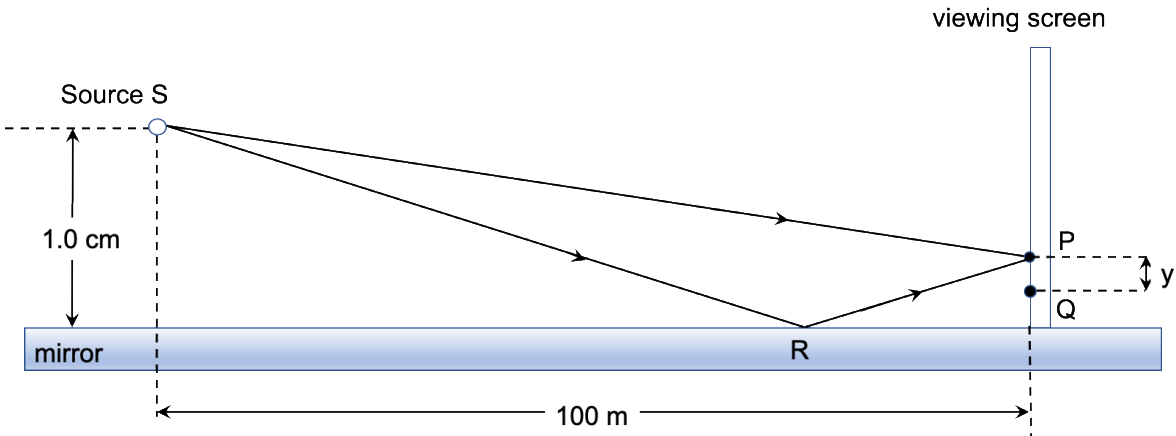


Fig. 3.1 (not to scale)

Points P and Q are at minimum intensity and there is only one point between P and Q which is at a maximum intensity.

S is placed 100 m to the left of the screen and 1.0 cm above the mirror.

	(i)	Explain why P is at a minimum intensity.	
			
			
			
			[1]
			1

			Solution:	
			The direct and reflected waves arrive at P in antiphase, undergoing destructive interference.	
		(ii)	Calculate the distance y. Explain your working.	
			y = mm	[3]
			y = fringe separation	
			$y = \frac{\lambda D}{a}$ $y = \frac{(500 \times 10^{-9})(100)}{2 \times (1.0 \times 10^{-2})}$ $= 2.5 \times 10^{-3} \text{ m} = 2.5 \text{ mm}$	1 1 1
		(iii)	Describe and explain the changes observed at point P, if any, as a result of the following changes made independently to the experiment.	
		1.	Source S is gradually shifted vertically further away from the mirror.	
				
				[3]
			As S is shifted away gradually, the PATH difference between the direct wave and the reflected wave arriving at P will increase continuously.	1
			This causes the PHASE difference between the waves at P to change continuously from 180° to 0°, back to 180° and so on.	1
			Thus alternating instances of constructive and destructive interferences will occur at P.	
			As a result, alternating instances of brightening and dimming of the light intensity will be observed at P.	1

			2.	The mirror is replaced by a black cardboard.	
					
					[2]
				Solution: The black cardboard is a very poor reflector of light (OR absorbs most of the light wave incident on it rather than reflect the wave). Thus interference between waves are no longer observable. P which was initially experience destructive interference will now appear brighter. [OR Thus the reflected wave will have significantly lower amplitude compared to the direct wave arriving at P. At P, destructive interference results in an intensity that is very similar to the intensity due to one wave only, so P appears brighter than before.]	1 1

4	(a)	(i)	State what is meant by the <i>internal energy</i> of a system
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